CDB update

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- Pixel masks
- detconfigs in CDB (reminder)
- cdbSummaryGT
- mcrealistic
- Global tags

Pixel masks

- Masked pixel will not readout a hit
  - this can (should) be taken into account in tracking
  - this information is available in (binary) mask files
    - on mu3ebe . . .
  - however
    - which mask loaded into detector hardware during which run
    - non-existent mask versioning
    - furthermore, additional "C++" masking (in addition to mask files)
  - this is not how "best practices" would look like
- Uploaded two sets of binary mask files to C(onfig)DB
  - tdac_files_bu_tune1_refined, from run 4763 onwards (9390 masked pixels)
  - tdac_files_bu_06_21_bestsofar, from run 5592 onwards (6814 masked pixels)
  - Two IOVs (plus one for "perfect" for start)

⇒ But the ConfigDB is not the CDB! ("detconfigs" collection in CDB)

- Notabene
  - large dataset 182MB = 2844*64kBit for full detector

calPixelMask

- calPixelQualityLM provides access to pixel quality
  - ▷ variety of sources (hitmaps, midas meta data, . . .)
  - ▷ keep this class as unified access to pixel quality information
    - masked status already foreseen there (and used, kind of)
    - backward compatibility

⇒ calPixelMask
- ▷ normal "calibration"/tag/IOVs in a GT
- ▷ payloads derived from binary files, stored in CDB(!)
- ▷ linked into calPixelQualityLM (with a pointer)
- ▷ backward compatible: if no calPixelMask present, ignore it
- ▷ access as before via calPixelQualityLM::getStatus(int chipid, int row, int col)

→ first available in
- mu3eUtil CDB-v6.6pre and (eventually) in v6.6
- GT datav6.6=2025V0

(TCS asked for this)

⇒ Need a bit more validation (still in "pre")

Reminder on detConfig up/download

- Query tags already in detconfigs collection

```
downloadDetConfig -l --host mu3edb0
```

- Uploading files from a directory and label them with a tag

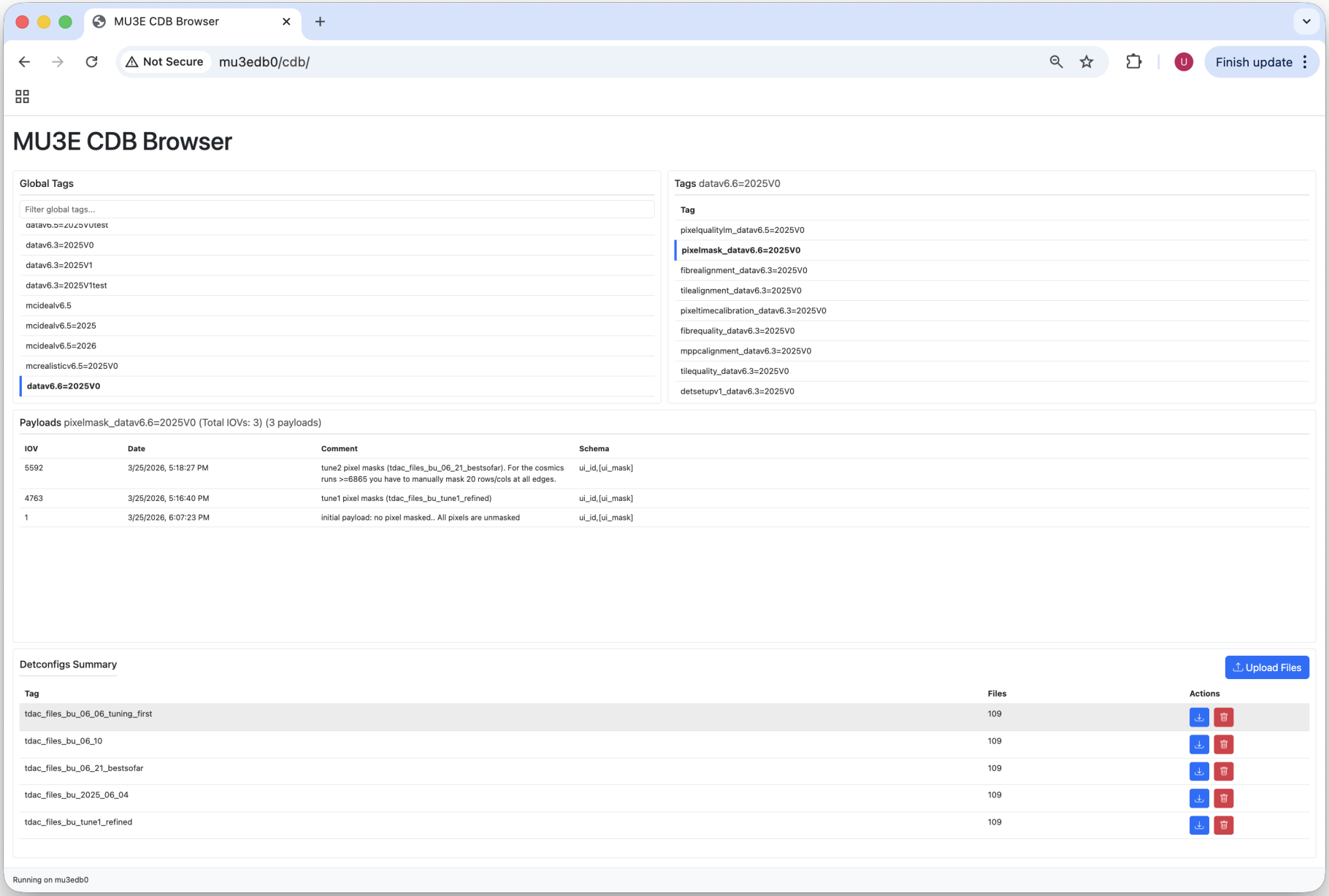
```
uploadDetConfig --host mu3edb0  \
            --dir  /Users/ursl/data/mu3e/masks/tdac_files_bu_06_21_bestsofar \
            --pattern mask_chip_  --tag tdac_files_bu_06_21_bestsofar
```

- Downloading files to a directory

```
downloadDetConfig --host mu3edb0 \
            --tag tdac_files_bu_06_21_bestsofar \
            --dir /Users/ursl/tmp/
```

- Note: also works via direct curl
  - in case you want to incorporate into your py scripts
  - see source code downloadDetConfig.cpp and uploadDetConfig.cpp

for the non-scripters



cdbSummaryGT

- Documentation for CDB contents needs improvement
  - now all GT and tags have documentation attached to them
  - could be read from within code
- Standalone executable for reading GT/tag documentation
  - mu3eUtil CDB-v6.6pre and (eventually) in v6.6

```
moor>./modules/mu3eUtil/cdb/test/cdbSummaryGT
Error: -u URI and -g GT are required
Usage: ./modules/mu3eUtil/cdb/test/cdbSummaryGT -u <URI> -g <GT>
  -u, --uri   CDB URI (path for JSON, http URL for REST)
  -g, --gt    Global tag name

moor>./modules/mu3eUtil/cdb/test/cdbSummaryGT -u ~/data/mu3e/cdb/ -g datav6.6=2025V0
cdbJSON::readGlobalTags() from  gtdir = /Users/ursl/data/mu3e/cdb//globaltags
GT: datav6.6=2025V0
  tags: 12
  comment: March 2026. Requires at least CDB-v6.6pre for mu3eUtil.
           Covers entire 2025 datataking period with two-layer VTX and
           fragmentary fibres and tiles (beam and cosmics, with proper
           B field). VTX alignment based on metrology and cosmics data
           (DF and MS, respectively). pixelQualityLM payloads correctly
           include LVDS errors from MIDAS meta data file and includes
           the pixel mask file information (except for cosmic runs >=
           6865, see pixelmask tag comment). pixelEfficiency with
           reduced iov list: 1 (perfect), 5700 (TCS, to be corrected)
```

  - Comments are "lightweight". If you think something is missing, tell me!
  - GTs (and the tags combined in them) are always "work in progress"

cdbSummaryGT (II)

```
 1) pixelalignment_datav6.3=2025V1
    IOV length: 1
    size: 108 detector units
    comment: VTX only, based on metrology and cosmics data (DF and MS,
             respectively)

 2) fibrealignment_datav6.3=2025V0
    IOV length: 1
    size: 4524 detector units
    comment: Database entry inserted

...

 5) pixelqualitylm_datav6.5=2025V0
    IOV length: 3330
    IOV     9361  size matches: OK
    all IOVs same size: OK
    agrees with alignment size
    comment: Correctly includes the LVDS errors from the midas meta data
             file

 6) pixelmask_datav6.6=2025V0
    IOV length: 3
    comment: pixel masks: Tune 1 and Tune 2. No proper mask files
             available for cosmic runs (additional masking of 20
             rows/cols around edges required - DIY)

 7) detsetupv1_datav6.3=2025V0
    IOV length: 3
    comment: Database entry inserted

...
```

mcrealisticv6.5=2025V0

- MC simulation in analysis
  - ▷ show that you quantitatively understand performance of your detector
  - ▷ show that you quantitatively understand your background
  - ▷ determine acceptance and efficiency
  - ▷ determine selection (optimize cuts, MVA training, . . .)
  - ▷ closure tests
  - → need proper simulation of resolution, time-walks, losses, noise, deficiencies, distortions, backgrounds
- This cannot be done in mu3eSim
  - ▷ "smear" the ideal MC simulation (every HEP experiment does it)
  - ▷ non-trivial (alignment, time-walk, defects, noise, . . .)

⇒ Started with a place-holder GT mcrealisticv6.5=2025V0

```
moor>./modules/mu3eUtil/cdb/test/cdbSummaryGT -u ~/data/mu3e/cdb/ -g mcrealisticv6.5=2025V0
cdbJSON::readGlobalTags() from  gtdir = /Users/ursl/data/mu3e/cdb//globaltags
GT: mcrealisticv6.5=2025V0
  tags: 11
  comment: MC realistic 2025 detector conditions with *placeholder* MC
           (no) smearing. Ideal 2025 geometries using MC truth from
           v6.5 (VTX + DS tiles + all fibres). Starting point for
           analysis of MC simulation data.
...
```

(Recent) Global tags in CDB

- We (= you) have some thinking/work to do
  - ▷ incorporate real-detector misalignment into realistic MC?
  - ▷ need smeared pixeltimecalibration time-walk calibration
  - ▷ does pixelefficiency belong into data GT? (no)
  - ▷ does it make sense to use a data GT for MC analysis? (no)

Global Tag (GT)	datav6.5=2025V0	datav6.6=2025V0	mcidealv6.5	mcrealisticv6.5=2025V0
mu3eUtil tag $\geq$	v6.5pre0	CDB-v6.6pre	v6.5pre0	v6.5pre0
pixelalignment	metrology/cosmics	ditto	ideal	ideal
fibrealignment	ideal	ditto	ideal	ideal
tilealignment	ideal	ditto	ideal	ideal
mppcalignment	ideal	ditto	ideal	ideal
pixelqualityLM	w/LVDS error rate	ditto + masks	ideal	as data
pixelmasks		yes		
fibrequality	99 IOVs	ditto	ideal	as data
tilequality	13 IOVs	ditto	ideal	as data
pixelefficiency	2 IOVs (biased)	ditto	ideal	as data
pixeltimecalibration	1 IOV	ditto	ideal	ideal
eventstuffv1	3552 IOV	ditto	ideal	ideal ?
detsetupv1	magnet off/on/off	ditto	on	on

Summary

- Pixel masks (to the extent they exist) incorporated into CDB
  - calPixelMask
- Documentation of GTs/tags
  - cdbSummaryGT
- CDB: nothing holds back tagging V6.5
  - (another) big feature branch CDB-v6.6pre growing/required for the above
- Outlook
  - payload browser for JSON backend